JEVIS: Joint Entropy Estimation in 3DGS via Visibility Field for Active Mapping

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***Abstract*— Active mapping requires planning observations that maximally reduce uncertainty about an unknown scene. Recent methods have substantially improved uncertainty quantification in 3D Gaussian Splatting (3DGS) by explicitly modeling visibility, assigning higher uncertainty to unobserved regions. However, these methods estimate the entropy of a single image observation and therefore primarily support myopic next-best-view selection. Practical deployments instead require estimating the joint entropy of multiple observations, such as views along a long-horizon trajectory or viewpoints assigned across a team of robots. In these settings, different views can observe common unseen regions of the scene, and the correlation between their observations is not negligible. To this end, we present Joint Entropy estimation in 3DGS via VISibility field (JEVIS), a theoretically grounded framework for quantifying the expected information gain of multiple planned observations in 3DGS through joint entropy estimation. By leveraging a maximum-likelihood observation (MLO) approximation, JEVIS efficiently estimates joint entropy across multiple views. We evaluate JEVIS in long-horizon single-agent and multi-agent active mapping across diverse indoor and space-robotics environments. Results show that JEVIS more accurately captures joint uncertainty and produces higher-fidelity reconstructions than existing methods. Project page: https://gatech-rl2.github.io/JEVIS/**

I. Introduction

Reliable autonomy in an unfamiliar environment requires a robot to reason about what it does not yet know about the scene. To reduce the uncertainty about the environment, the robot must actively explore and understand its surroundings, a process known as active mapping [1]–[3]. Active mapping is typically formulated as the problem of planning actions that maximize expected information gain, or equivalently, minimize uncertainty about the reconstructed scene [4]–[6]. It is essential for applications including household, search-and-rescue, and space robotics. Achieving this objective requires a scene representation that supports both high-quality reconstruction and reliable uncertainty quantification. Recently, radiance fields [7], [8] have attracted widespread interest due to their high-fidelity reconstruction and fast rendering capabilities, and have been increasingly adopted as representations for active mapping [5], [6], [9]. However, accurately and efficiently quantifying uncertainty in radiance fields remains challenging due to their large number of trainable parameters [9], [10].

Recent methods have substantially advanced uncertainty quantification in radiance fields for active mapping by explicitly modeling visibility [5], [6], [11], motivated by the intuition that predictions for regions unseen in the training views should carry higher uncertainty, but their formulations typically quantify uncertainty for only one image observation at a time. In practical deployments, robots may need to plan several actions ahead for non-myopic and more efficient exploration, or work in teams to accelerate mapping through parallel exploration. However, uncertainty estimates for individual image observations alone are insufficient to evaluate such plans, limiting the use of existing methods in long-horizon and multi-agent active mapping.

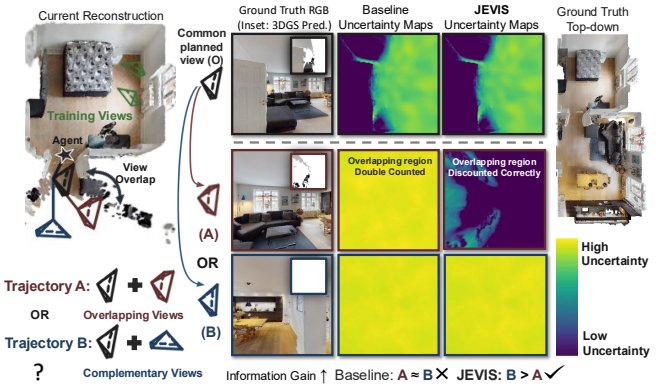


Fig. 1. **JEVIS estimates the joint entropy of multiple observations along a trajectory by modeling their dependencies.** A robot is exploring a partially reconstructed scene: the upper room is fully reconstructed, while the lower room remains unexplored. The unreliable reconstruction of the unexplored room produces largely incomplete 3DGS renderings (insets). The robot plans a trajectory to maximize expected information gain and thereby improve reconstruction of the entire scene. Two candidate trajectories leading into the unexplored region share the first planned view O, then continue to either A, which reobserves the same region, or B, which observes a different region. Treating observations as independent [6] double counts uncertainty in overlapping regions, incorrectly assigning similar information-gain estimates to both trajectories. By accounting for this redundancy, JEVIS correctly favors the more informative trajectory $O \to B$ over $O \to A$.

Our key insight is that multiple planned views may observe the same regions unseen in the training data and therefore provide partially redundant information. Naively summing per-image entropies counts uncertainty in these shared unseen regions repeatedly, overestimating the joint uncertainty of the planned observations. Accurately estimating their joint entropy is therefore crucial for information-theoretic planning in long-horizon and multi-agent active mapping.

In this work we present **Joint Entropy estimation in 3DGS via VISibility field (JEVIS)**, a theoretically grounded framework for quantifying the expected information gain of multiple planned observations in 3DGS through joint entropy estimation. JEVIS uses Visibility Fields [6] to model uncertainty shared by planned views of common regions unseen in the training data. We prove that JEVIS matches the

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entropy estimates of an explicit maximum-likelihood observation procedure while avoiding repeated 3DGS retraining.

To summarize, our main contributions are:

- We establish a theoretically grounded framework for quantifying the expected information gain of multiple planned observations in 3DGS through joint entropy estimation, enabling long-horizon active mapping.
- We extend this formulation to multi-robot active mapping, providing, to our knowledge, the first principled information-theoretic treatment of long-horizon multi-agent active mapping with radiance fields.
- We evaluate JEVIS across diverse indoor and space-robotics environments, demonstrating consistently higher reconstruction quality than existing baselines in both single- and multi-agent settings.

II. Related Work

Active mapping methods can be broadly categorized into approaches based on hand-crafted heuristics [1], learned policies [12], or information-theoretic objectives [13]. The first two guide planning with limited theoretical guarantees and uncertain generalization to unseen environments. In contrast, information-theoretic methods quantify uncertainty in scene representations or future observations to select trajectories that maximize expected information gain [13], [14]. Traditional map representations include sparse landmarks [15], [16] and occupancy grids [14], whose reconstruction detail is limited by sparsity and spatial discretization, respectively. Since NeRF [7], radiance fields have advanced dense 3D reconstruction; 3D Gaussian Splatting (3DGS) [8] enables fast, high-fidelity rendering and is widely used in SLAM [17], [18]. We focus on information-theoretic active mapping with 3DGS representations in this work.

a) Radiance Field Uncertainty Quantification: Epistemic uncertainty quantification in radiance fields guides active mapping toward observations that improve reconstruction quality [19]. One line of work directly adapts Bayesian uncertainty estimation techniques used in active learning to radiance fields. Bayes' Rays [20] and FisherRF [9] use Laplace approximations over spatial perturbations and radiance-field parameters, respectively; FisherRF uses Fisher information to approximate expected information gain for view selection. VIMC [10] instead models parameter uncertainty through variational sampling on a low-dimensional manifold. However, applying these techniques to high-dimensional radiance fields with strongly correlated parameters relies on restrictive approximations, which can yield uncertainty estimates that poorly reflect reconstruction error and limit active mapping performance [6]. Recent visibility-based methods have advanced the state of the art in uncertainty quantification and active view selection, motivated by the observation that regions unseen during training yield unreliable predictions. NVF [5] integrates visibility into a Bayesian network for uncertainty-aware NeRF volume rendering, while GAVIS [6] extends this formulation to 3DGS with anisotropic visibility. COVER [11] derives a coverage-based surrogate for Fisher information gain that prioritizes insufficiently observed regions. These methods support myopic next-best-view planning, but their per-view objectives ignore dependence among future observations; summing these scores along a trajectory can therefore overcount information in overlapping views.

b) Long-Horizon Active Mapping: Long-horizon active mapping plans trajectories to maximize reconstruction quality [4], [21], [22]. Traditional information-theoretic methods evaluate trajectories by the expected information gain of future observations [13], [14]. The maximum-likelihood observation (MLO) approximation avoids integrating over all possible observation sequences by propagating a nominal belief trajectory using the most likely measurement at each step [23]. However, traditional uncertainty models do not readily scale from sparse landmarks [22] or occupancy grids [14] to radiance fields with large, correlated parameter spaces. Although FisherRF [9] supports batch acquisition, its unreliable uncertainty estimates in unseen regions [6] can impair trajectory evaluation. Heuristic methods instead use rendering-based gains [24] or topological guidance [25]. In particular, MAGICIAN [26] maximizes long-horizon surface coverage using imagined 3D Gaussians in observed and unseen regions, but its generalization may be constrained by its pretrained occupancy network.

c) Multi-Agent Active Mapping: In the multi-agent mapping setting, observation redundancy arises both within individual trajectories and across trajectories committed by different robots. Classical approaches include sequential path allocation by residual information gain under Gaussian-process models [27] and joint robot–landmark entropy minimization [28]. Heuristic alternatives include frontier exploration [29] and semantic-guided 3DGS reconstruction using surface-uncertainty scores [30]. However, to our knowledge, no information-theoretic framework yet quantifies joint information gain across multiple trajectories for long-horizon multi-agent active mapping with 3DGS.

III. Background

This section reviews the foundations of JEVIS: 3DGS for scene representation, visibility-aware volume rendering for uncertainty quantification, and GAVIS's anisotropic visibility estimator. Further details are available in [5], [6], [8].

a) 3D Gaussian Splatting: 3DGS [8] represents a scene as a set of anisotropic Gaussian particles $\{\mathcal{G}_i\}$, each parameterized by a mean $\boldsymbol{\mu}_i \in \mathbb{R}^3$, covariance $\boldsymbol{\Sigma}_i$, opacity, and view-dependent color $\boldsymbol{c}_i(\boldsymbol{d})$ encoded with spherical harmonics (SH). To render a pixel along ray $\boldsymbol{r}$ with direction $\boldsymbol{d}$, the particles intersecting the ray are sorted front-to-back and their contributions are alpha-composited,

$$\boldsymbol{C}(\boldsymbol{r}) = \sum_i w_i \, \boldsymbol{c}_i(\boldsymbol{d}), \qquad w_i = \alpha_i \prod_{j<i}(1-\alpha_j), \quad (1)$$

where α_i is particle opacity times the projected Gaussian value at the pixel and $T_i = \prod_{j<i}(1-\alpha_j)$ is transmittance to particle i. Compared with NeRF [7], this explicit and differentiable rasterization yields both faster training and real-time rendering, though the number of trainable parameters

scales with the number of primitives, making uncertainty quantification expensive.

b) Uncertainty-aware Volume Rendering via Visibility Field: NVF [5] incorporates visibility into a Bayesian-network formulation of uncertainty-aware volume rendering to account for unreliable color predictions in unseen regions. It models each particle's emission color as a Gaussian and represents the rendered pixel-color distribution as a Gaussian mixture model (GMM), with mixture weights derived from alpha compositing. The visibility $v_i \in [0,1]$ denotes the probability that particle i has been observed by at least one training view. Weighting the learned color distribution by v_i and a high-variance prior for unseen regions by $1 - v_i$ gives

$$p(\boldsymbol{z}) = \textstyle\sum_i w_i^* \, v_i \mathcal{N}(\boldsymbol{\mu}_{\boldsymbol{c}_i}, \boldsymbol{Q}_{\boldsymbol{c}_i}) + \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{Q}_0) \textstyle\sum_i w_i^* (1 - v_i), \quad (2)$$

where w_i^* is the visibility-corrected mixture weight and $\mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{Q}_0)$ is a high-variance prior assigned to unseen regions. The pixel entropy $\mathcal{H}(\boldsymbol{Z})$ used for active mapping is then computed from (2) with an upper-bound GMM entropy estimator [31]. NVF approximates $v_i = V_\theta(\boldsymbol{x}_i)$ with an auxiliary neural network that takes only position as input, which is slow to train and cannot capture the anisotropy in visibility arising from self-occlusion.

c) Anisotropic Visibility Field for 3DGS: GAVIS [6] replaces the neural network of NVF with an analytical, direction-aware construction for 3DGS, yielding a per-particle visibility $V^{(i)}(\boldsymbol{d})$ depending on the rendering direction $\boldsymbol{d}$. For one training pose $\boldsymbol{p} \in \mathcal{P}$ with viewing direction $\boldsymbol{d}_p$, the single-view visibility of particle i factorizes as

$$V_{\boldsymbol{p}}^{(i)}(\boldsymbol{d}) \;=\; \Phi_{i,\boldsymbol{p}} \, T_{\boldsymbol{p}}(t_i^{\boldsymbol{p}}) \, \nu(\boldsymbol{d}; \boldsymbol{d}_p), \quad (3)$$

where $\Phi_{i,\boldsymbol{p}} \in \{0,1\}$ indicates whether particle i lies within camera $\boldsymbol{p}$'s field of view, $T_{\boldsymbol{p}}(t_i^{\boldsymbol{p}})$ is the transmittance from camera $\boldsymbol{p}$ to particle i along $\boldsymbol{d}_p$ obtained directly from the rasterizer, and $\nu(\boldsymbol{d}; \boldsymbol{d}_p) = \exp\big(\kappa(\boldsymbol{d} \cdot \boldsymbol{d}_p - 1)\big)$ is a von Mises-Fisher-shaped directional kernel that decays as the query direction $\boldsymbol{d}$ deviates from $\boldsymbol{d}_p$. The per-particle visibility with respect to the entire training set is then the probability of the particle being visible from at least one training view,

$$V^{(i)}(\boldsymbol{d}) \;=\; 1 - \prod_{\boldsymbol{p} \in \mathcal{P}} \big(1 - V_{\boldsymbol{p}}^{(i)}(\boldsymbol{d})\big). \quad (4)$$

Evaluating (4) naively scales with $|\mathcal{P}|$ at query time, which is impractical for robot planning. GAVIS instead expands the per-view visibility in SH and stores, for each particle, a constant-size set of SH coefficients $\{\gamma_{\ell m}^{\mathcal{P}}\}$ aggregated across all training poses, so that an AM-GM lower bound on (4) can be queried in constant time independent of $|\mathcal{P}|$. The resulting visibility field can be built in under a second from a trained 3DGS and supplies the per-particle $v_i = V^{(i)}(\boldsymbol{d})$ that drives the GMM in (2).

NVF and GAVIS quantify single-image uncertainty but do not account for dependencies among multiple planned observations. Building on this line of work, JEVIS provides a theoretically grounded framework for estimating the joint entropy of multiple observations.

IV. Method

This section introduces JEVIS, our framework for quantifying the joint uncertainty of a set of observations given a trained 3DGS reconstruction, and describes how it can be applied to long-horizon and multi-agent active mapping. We first formulate long-horizon active mapping as joint entropy maximization (Sec. IV-A) and outline an explicit estimator based on the maximum-likelihood observation approximation (Sec. IV-B). We then present the efficient JEVIS estimator (Sec. IV-C) and establish its equivalence to the explicit procedure (Sec. IV-D). Finally, we describe how JEVIS supports multi-agent active mapping through sequential greedy trajectory assignment (Sec. IV-E).

A. Long-Horizon Active Mapping

Long-horizon active mapping selects a sequence of future camera poses that maximizes their joint expected information gain (EIG) about the scene. At time t, let $\boldsymbol{p}_{t+1:t+L} = (\boldsymbol{p}_{t+1}, \ldots, \boldsymbol{p}_{t+L})$ denote a candidate pose sequence over a planning horizon of length L. Such a sequence may correspond to a planned trajectory for a single robot or a team of robots. The next-best-view problem is recovered as the special case $L = 1$. Let $\boldsymbol{Z}_k$ denote the rendered observation at pose $\boldsymbol{p}_k$, and write $\boldsymbol{Z}_{t+1:t+L} = (\boldsymbol{Z}_{t+1}, \ldots, \boldsymbol{Z}_{t+L})$ for the corresponding sequence of observations; here we drop the explicit dependence on $\boldsymbol{p}_k$ to keep the notation compact. The EIG is then

$$\begin{aligned} I(\boldsymbol{Z}_{t+1:t+L}; \boldsymbol{\theta} \mid \boldsymbol{p}_{t+1:t+L}) = \mathcal{H}[\boldsymbol{Z}_{t+1:t+L} \mid \boldsymbol{p}_{t+1:t+L}] \\ - \mathcal{H}[\boldsymbol{Z}_{t+1:t+L} \mid \boldsymbol{\theta}, \boldsymbol{p}_{t+1:t+L}], \end{aligned} \quad (5)$$

where $I(\cdot)$ denotes mutual information, $\mathcal{H}[\cdot]$ is entropy, and $\boldsymbol{\theta}$ represents the radiance-field parameters.

We assume a Gaussian observation model that is conditionally independent across poses given $\boldsymbol{\theta}$, specifically,

$$p(\boldsymbol{Z}_{t+1:t+L} \mid \boldsymbol{p}_{t+1:t+L}, \boldsymbol{\theta}) = \prod_{k=t+1}^{t+L} \mathcal{N}(f(\boldsymbol{p}_k; \boldsymbol{\theta}), \, \boldsymbol{\Sigma}_z), \quad (6)$$

where $f(\boldsymbol{p}_k; \boldsymbol{\theta})$ is the rendering function and $\boldsymbol{\Sigma}_z$ is the observation noise covariance. Assuming $\boldsymbol{\Sigma}_z$ is constant and isotropic, i.e., $\boldsymbol{\Sigma}_z = \sigma_z^2 \mathbf{I}$, the conditional entropy reduces to $\mathcal{H}[\boldsymbol{Z}_{t+1:t+L} \mid \boldsymbol{\theta}, \boldsymbol{p}_{t+1:t+L}] = \sum_{k=t+1}^{t+L} \mathcal{H}[\boldsymbol{Z}_k \mid \boldsymbol{\theta}, \boldsymbol{p}_k]$, which is constant in $\boldsymbol{p}_{t+1:t+L}$ and can be omitted during optimization. Maximizing EIG therefore reduces to maximizing the joint entropy of the observations, which by the chain rule of entropy decomposes into a sum of conditional entropies:

$$\begin{aligned} \boldsymbol{p}_{t+1:t+L}^* &= \operatorname*{arg\,max}_{\boldsymbol{p}_{t+1:t+L}} \; \mathcal{H}(\boldsymbol{Z}_{t+1:t+L}) \\ &= \operatorname*{arg\,max}_{\boldsymbol{p}_{t+1:t+L}} \; \sum_{k=t+1}^{t+L} \mathcal{H}(\boldsymbol{Z}_k \mid \boldsymbol{Z}_{t+1:k-1}), \end{aligned} \quad (7)$$

with the convention that $\boldsymbol{Z}_{t+1:t}$ is the empty set, so the first term is the marginal entropy $\mathcal{H}(\boldsymbol{Z}_{t+1})$. Each conditional entropy $\mathcal{H}(\boldsymbol{Z}_k \mid \boldsymbol{Z}_{t+1:k-1})$ measures only the *additional* information that view k contributes beyond the views already planned at steps $t+1, \ldots, k-1$, naturally accounting

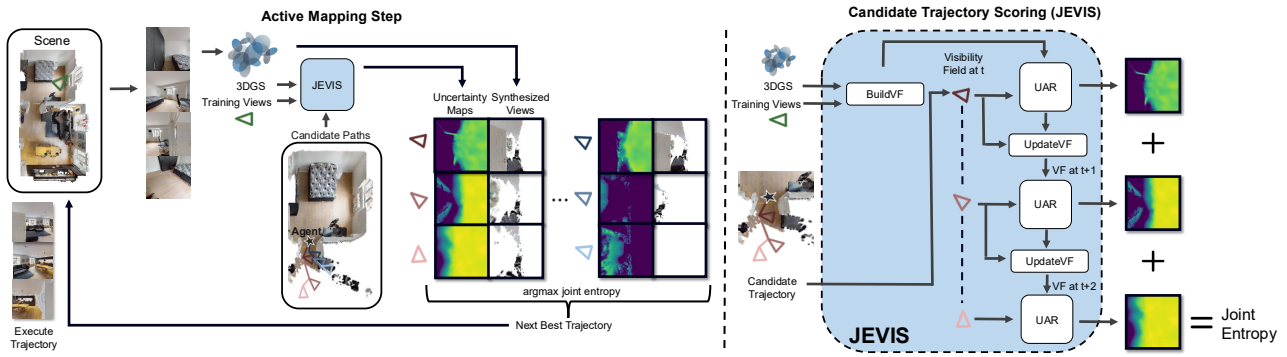


Fig. 2. **Active mapping with JEVIS.** (Left, Sec. IV-A) JEVIS scores candidate trajectories by their estimated joint observation entropy. The agent executes the highest-scoring trajectory and updates the 3DGS reconstruction with the acquired observations. (Right, Sec. IV-C) JEVIS first builds a visibility field (VF) from the trained 3DGS model and training poses. For each candidate trajectory, uncertainty aware 3DGS rasterization (UAR) estimates the first view's entropy using this initial VF. Before each subsequent view is evaluated, `UpdateVF` incorporates the previous pose into the VF, and UAR estimates the current view's entropy conditioned on all preceding views along the trajectory. These cumulative updates account for uncertainty shared across candidate views of regions unseen in the training views. Summing the first view's entropy and subsequent conditional entropies yields the trajectory's joint entropy estimate. Three views are shown for illustration; JEVIS supports trajectories of arbitrary length.

for redundancy between overlapping views. Consequently, $\mathcal{H}(\boldsymbol{Z}_{t+1:t+L})$ is no greater than the sum of per-view marginal entropies $\sum_{k=t+1}^{t+L} \mathcal{H}(\boldsymbol{Z}_k)$, since overlapping views share information through the common scene parameters $\boldsymbol{\theta}$.

At each planning iteration, we fit a 3DGS to the acquired observations, sample candidate trajectories, and execute the one with the highest estimated joint entropy (Fig. 2, left). The newly acquired observations are added to the training set for the next planning iteration.

B. Maximum-Likelihood Observation Approximation

Evaluating each conditional entropy term in (7) exactly requires marginalizing over the unknown realizations of the planned but not yet acquired observations $\boldsymbol{Z}_{t+1:k-1}$, which is intractable for radiance fields with millions of parameters. We therefore adopt the *maximum-likelihood observation* (MLO) approximation that replaces these random variables with their most likely values under the current model. This approximation has been used in belief-space planning [23] and active SLAM [13] to obtain a tractable nominal belief trajectory. From initial 3DGS parameters $\boldsymbol{\theta}_{\text{init}}$ (the previous reconstruction when available), `Optimize` minimizes $\mathcal{L}_t(\boldsymbol{\theta}) = \sum_{j=1}^{t} \|f(\boldsymbol{p}_j;\boldsymbol{\theta}) - \boldsymbol{z}_j\|_2^2$ to obtain $\boldsymbol{\theta}_t^*$. Under our Gaussian observation model, we use its rendered mean as the MLO surrogate, $\hat{\boldsymbol{z}}_j = f(\boldsymbol{p}_j;\boldsymbol{\theta}_t^*)$, giving

$$\mathcal{H}(\boldsymbol{Z}_k \mid \boldsymbol{Z}_{t+1:k-1}) \approx \mathcal{H}\big(\boldsymbol{Z}_k \,\big|\, \boldsymbol{Z}_{t+1:k-1} = \hat{\boldsymbol{z}}_{t+1:k-1}\big), \quad (8)$$

where $\hat{\boldsymbol{z}}_{t+1:k-1} \triangleq (\hat{\boldsymbol{z}}_{t+1}, \ldots, \hat{\boldsymbol{z}}_{k-1})$ is fixed throughout candidate evaluation. This collapses an intractable marginalization to a single point evaluation while preserving the cross-view dependence on $\boldsymbol{\theta}$.

We first describe a straightforward but costly explicit MLO procedure for estimating the joint entropy of a candidate trajectory. Given a trained 3DGS and a pose sequence, `BuildVF` constructs the anisotropic visibility field (VF) described in Sec. III by applying `UpdateVF` to each pose. Following GAVIS [6], `UpdateVF` accumulates each pose's SH visibility contributions to update the VF. The explicit MLO procedure starts from $\boldsymbol{\theta}_t^*$ and uses warm-started incremental optimization as the MLO surrogates of preceding planned views are added to the acquired observations. Before each query, it rebuilds the VF from the acquired and preceding planned poses under the resulting model, similar to the VF construction procedure of GAVIS. The uncertainty-aware 3DGS rasterization (`UAR`) routine then uses the VF and 3DGS model to estimate the current view's conditional entropy following (2); summing these estimates yields the trajectory's joint entropy estimate via (7) (see Sec. IV-D for details). Repeating 3DGS optimization and VF construction along every candidate trajectory is computationally prohibitive for online planning.

C. Joint Entropy Estimation via Visibility Field

JEVIS avoids repeated 3DGS optimization and VF construction by keeping the trained model $\boldsymbol{\theta}_t^*$ fixed and incrementally updating the VF along each candidate trajectory (Fig. 2, right; Alg. 1). Let $\mathcal{V}_k$ denote the VF after incorporating poses $\boldsymbol{p}_{1:k}$ under the fixed model $\boldsymbol{\theta}_t^*$, for $k \geq t$. Each candidate trajectory is evaluated starting from $\mathcal{V}_t = \texttt{BuildVF}(\boldsymbol{\theta}_t^*, \boldsymbol{p}_{1:t})$, the VF constructed from the training poses. At each candidate pose $\boldsymbol{p}_k$, we estimate an entropy map from the GMM in (2) using $\mathcal{V}_{k-1}$ and the fixed 3DGS parameters $\boldsymbol{\theta}_t^*$. Following [5] and [6], we compute the per-pose entropy $\mathcal{H}_k$ from pixel entropies. It estimates $\mathcal{H}(\boldsymbol{Z}_k \mid \boldsymbol{Z}_{t+1:k-1})$ under the MLO approximation (8). We then update the field as $\mathcal{V}_k = \texttt{UpdateVF}(\mathcal{V}_{k-1}, \boldsymbol{\theta}_t^*, \boldsymbol{p}_k)$ before evaluating the next pose. Summing these conditional entropy estimates yields the trajectory's estimated joint entropy via the chain-rule decomposition in (7). Without these updates, the field would remain fixed at $\mathcal{V}_t$, reducing the trajectory score to a sum of per-view marginal entropies that overcounts information shared by overlapping views. In Sec. IV-D, we prove that JEVIS and the explicit MLO procedure yield the same entropy estimates under the stated assumptions.

D. Equivalence to the Explicit MLO Procedure

To establish equivalence, we formalize the explicit MLO procedure outlined in Sec. IV-B (Alg. 2). For step k, the

Algorithm 1 JEVIS: Joint Entropy Estimation

Input: Initial 3DGS parameters $\boldsymbol{\theta}_{\text{init}}$, past observations $\boldsymbol{z}_{1:t}$, past poses $\boldsymbol{p}_{1:t}$, candidate trajectory $\boldsymbol{p}_{t+1:t+L}$
Output: Joint entropy $\mathcal{H}$

▷ *Train 3DGS*
$\boldsymbol{\theta}_t^* \leftarrow \texttt{Optimize}(\mathcal{L}_t; \boldsymbol{\theta}_{\text{init}})$
▷ *Visibility field construction from training poses*
$\mathcal{V}_t \leftarrow \texttt{BuildVF}(\boldsymbol{\theta}_t^*, \boldsymbol{p}_{1:t})$
▷ *Joint entropy estimation along the candidate trajectory*
for $k = t+1$ **to** $t+L$ **do**
  $\mathcal{H}_k \leftarrow \texttt{UAR}(\boldsymbol{p}_k; \mathcal{V}_{k-1}, \boldsymbol{\theta}_t^*)$
  $\mathcal{V}_k \leftarrow \texttt{UpdateVF}(\mathcal{V}_{k-1}, \boldsymbol{\theta}_t^*, \boldsymbol{p}_k)$
end for
return $\sum_{k=t+1}^{t+L} \mathcal{H}_k$

augmented reconstruction loss is

$$\mathcal{L}_{k-1}(\boldsymbol{\theta}) = \mathcal{L}_t(\boldsymbol{\theta}) + \sum_{j=t+1}^{k-1} \|f(\boldsymbol{p}_j; \boldsymbol{\theta}) - \hat{\boldsymbol{z}}_j\|_2^2, \tag{9}$$

where $\hat{\boldsymbol{z}}_j = f(\boldsymbol{p}_j; \boldsymbol{\theta}_t^*)$ are the fixed MLO surrogates. The explicit procedure reuses $\boldsymbol{\theta}_t^*$ at $k = t+1$ and, for $k \geq t+2$, optimizes $\mathcal{L}_{k-1}$, initialized from the preceding solution $\boldsymbol{\theta}_{k-2}^*$. Let $\mathcal{W}_j^{(k)}$ denote the VF incorporating poses $\boldsymbol{p}_{1:j}$ under $\boldsymbol{\theta}_{k-1}^*$, for $t \leq j < k$. Before querying $\boldsymbol{p}_k$, Alg. 2 rebuilds the VF from the training poses $\boldsymbol{p}_{1:t}$ under $\boldsymbol{\theta}_{k-1}^*$, then incorporates the preceding planned poses $\boldsymbol{p}_{t+1:k-1}$.

Algorithm 2 Explicit MLO Procedure (for Reference)

Input: Initial 3DGS parameters $\boldsymbol{\theta}_{\text{init}}$, past observations $\boldsymbol{z}_{1:t}$, past poses $\boldsymbol{p}_{1:t}$, candidate trajectory $\boldsymbol{p}_{t+1:t+L}$
Output: Joint entropy $\mathcal{H}$

▷ *Train 3DGS*
$\boldsymbol{\theta}_t^* \leftarrow \texttt{Optimize}(\mathcal{L}_t; \boldsymbol{\theta}_{\text{init}})$
for $k = t+1$ **to** $t+L$ **do**
  if $k \geq t+2$ **then**
    $\boldsymbol{\theta}_{k-1}^* \leftarrow \texttt{Optimize}(\mathcal{L}_{k-1}; \boldsymbol{\theta}_{k-2}^*)$ ▷ warm start
  end if
  $\mathcal{W}_t^{(k)} \leftarrow \texttt{BuildVF}(\boldsymbol{\theta}_{k-1}^*, \boldsymbol{p}_{1:t})$
  for $j = t+1$ **to** $k-1$ **do**
    $\mathcal{W}_j^{(k)} \leftarrow \texttt{UpdateVF}(\mathcal{W}_{j-1}^{(k)}, \boldsymbol{\theta}_{k-1}^*, \boldsymbol{p}_j)$
  end for
  $\mathcal{H}'_k \leftarrow \texttt{UAR}(\boldsymbol{p}_k; \mathcal{W}_{k-1}^{(k)}, \boldsymbol{\theta}_{k-1}^*)$
end for
return $\sum_{k=t+1}^{t+L} \mathcal{H}'_k$

The idealized `Optimize` routine returns a local minimizer and retains its initialization whenever that initialization is already a local minimizer of the input loss.

Proposition 1 (Equivalence under MLO): Suppose both algorithms share the fitted model $\boldsymbol{\theta}_t^*$, a local minimizer of $\mathcal{L}_t$, and use the same deterministic `BuildVF`, `UpdateVF`, and `UAR` routines. Under the MLO approximation (8) and the optimization rule above, the per-step entropy estimates from Alg. 1 and Alg. 2 satisfy $\mathcal{H}_k = \mathcal{H}'_k$ for every $k \in \{t+1, \ldots, t+L\}$, and consequently the two algorithms return the same joint entropy estimate $\sum_{k=t+1}^{t+L} \mathcal{H}_k = \sum_{k=t+1}^{t+L} \mathcal{H}'_k$.

Proof: (i) The model parameters do not change: $\boldsymbol{\theta}_{k-1}^* = \boldsymbol{\theta}_t^*$ for all $k \in \{t+1, \ldots, t+L\}$. The added terms in (9) are nonnegative and vanish at $\boldsymbol{\theta}_t^*$. Since $\boldsymbol{\theta}_t^*$ locally minimizes $\mathcal{L}_t$, for every $\boldsymbol{\theta}$ in a neighborhood of $\boldsymbol{\theta}_t^*$ we have

$$\mathcal{L}_{k-1}(\boldsymbol{\theta}) \geq \mathcal{L}_t(\boldsymbol{\theta}) \geq \mathcal{L}_t(\boldsymbol{\theta}_t^*) = \mathcal{L}_{k-1}(\boldsymbol{\theta}_t^*). \tag{10}$$

Thus $\boldsymbol{\theta}_t^*$ remains a local minimizer of every augmented loss $\mathcal{L}_{k-1}$. The first step reuses $\boldsymbol{\theta}_t^*$. At each subsequent step, if $\boldsymbol{\theta}_{k-2}^* = \boldsymbol{\theta}_t^*$, then `Optimize` starts at a local minimizer and returns it unchanged. Induction therefore gives $\boldsymbol{\theta}_{k-1}^* = \boldsymbol{\theta}_t^*$ for all candidate steps.

(ii) The query visibility fields agree: $\mathcal{W}_{k-1}^{(k)} = \mathcal{V}_{k-1}$. Fix any candidate step k. By claim (i) and the same VF construction procedure, $\mathcal{W}_t^{(k)} = \texttt{BuildVF}(\boldsymbol{\theta}_t^*, \boldsymbol{p}_{1:t}) = \mathcal{V}_t$. For each $j = t+1, \ldots, k-1$, both constructions apply the same deterministic `UpdateVF` routine with model $\boldsymbol{\theta}_t^*$ and pose $\boldsymbol{p}_j$. Induction over j gives $\mathcal{W}_j^{(k)} = \mathcal{V}_j$, including all field state used by the entropy query, and hence $\mathcal{W}_{k-1}^{(k)} = \mathcal{V}_{k-1}$.

(iii) The per-step entropies agree: $\mathcal{H}'_k = \mathcal{H}_k$. By claim (i), $\boldsymbol{\theta}_{k-1}^* = \boldsymbol{\theta}_t^*$; by claim (ii), $\mathcal{W}_{k-1}^{(k)} = \mathcal{V}_{k-1}$. Substituting into the UAR query,

$$\mathcal{H}'_k = \texttt{UAR}(\boldsymbol{p}_k; \mathcal{W}_{k-1}^{(k)}, \boldsymbol{\theta}_{k-1}^*) = \texttt{UAR}(\boldsymbol{p}_k; \mathcal{V}_{k-1}, \boldsymbol{\theta}_t^*) = \mathcal{H}_k. \tag{11}$$

Summing over all $k = t+1, \ldots, t+L$ yields the claimed equality of joint entropy estimates. ■

The equivalence also holds for a finite-tolerance implementation of `Optimize` that checks its initialization before taking any update and returns it unchanged whenever the input loss is below a fixed tolerance ϵ. If the shared fitted model satisfies $\mathcal{L}_t(\boldsymbol{\theta}_t^*) < \epsilon$, then $\mathcal{L}_{k-1}(\boldsymbol{\theta}_t^*) = \mathcal{L}_t(\boldsymbol{\theta}_t^*) < \epsilon$, since all added MLO residuals vanish. Every warm-started call therefore returns the same model, and parts (ii)–(iii) of the proof apply unchanged. Thus, this implementation yields the same entropy estimates as JEVIS without requiring the initial fit to be an exact local minimizer.

E. Multi-agent Active Mapping

In the multi-agent setting, M agents must each commit to one trajectory drawn from a per-agent candidate pool $\mathcal{P}^{(m)}$, and we seek an assignment $\{\boldsymbol{p}_{t+1:t+L}^{(m)*}\}_{m=1}^{M}$ that maximizes the joint information gain over all $M \cdot L$ planned views. Exhaustive evaluation over the product space $\mathcal{P}^{(1)} \times \cdots \times \mathcal{P}^{(M)}$ is exponential in M and rapidly becomes intractable. Under our observation model (6), EIG is monotone submodular, so exact greedy selection under the one-trajectory-per-agent partition matroid achieves at least $1/2$ of the optimal EIG over the candidate pools [32]. Accordingly, we use JEVIS estimates to guide sequential greedy trajectory assignment.

Sequential greedy trajectory assignment uses a shared visibility field so that each agent's candidate trajectory scores account for observations already planned by earlier agents. As shown in Alg. 3, we maintain a shared visibility field $\mathcal{V}$,

Algorithm 3 Multi-Agent Active Mapping

Input: Per-agent candidate pools $\{\mathcal{P}^{(m)}\}_{m=1}^{M}$, trained 3DGS parameters $\boldsymbol{\theta}_t^*$, visibility field $\mathcal{V}_t$
Output: Multi-agent trajectory assignment $\{\boldsymbol{p}_{t+1:t+L}^{(m)*}\}_{m=1}^{M}$

$\mathcal{V} \leftarrow \mathcal{V}_t$
for $m = 1$ **to** M **do**
  $\boldsymbol{p}_{t+1:t+L}^{(m)*} \leftarrow \arg\max_{\boldsymbol{p} \in \mathcal{P}^{(m)}} \texttt{JEVIS}(\boldsymbol{p}; \mathcal{V}, \boldsymbol{\theta}_t^*)$
  for $k = t+1$ **to** $t+L$ **do**
    $\mathcal{V} \leftarrow \texttt{UpdateVF}(\mathcal{V}, \boldsymbol{\theta}_t^*, \boldsymbol{p}_k^{(m)*})$
  end for
end for
return $\{\boldsymbol{p}_{t+1:t+L}^{(m)*}\}_{m=1}^{M}$

initialized to $\mathcal{V}_t$, and visit agents sequentially from $m = 1$ to M. For each agent m, we select the trajectory in $\mathcal{P}^{(m)}$ that maximizes its JEVIS score against $\mathcal{V}$, then absorb each of its L poses into $\mathcal{V}$ via `UpdateVF` before processing the next agent. The same visibility-update mechanism thus discounts redundancy both within trajectories and across agents.

V. Experiments

We perform experiments to address the following questions: (**Q1**) How accurately and efficiently can JEVIS quantify the joint uncertainty of multiple synthesized views of 3DGS? (**Q2**) Can JEVIS enable effective long-horizon single-agent active mapping? (**Q3**) Can JEVIS enable effective long-horizon multi-agent active mapping?

A. Experiment Setup

a) Environments: We evaluate on 8 Gibson [33] and 16 Habitat-Matterport 3D (HM3D) [34] indoor scenes, plus 6 space scenes following [6]. The HM3D scenes are split by floor area into 8 standard-scale (HM3D-small) and 8 large-scale (HM3D-large) scenes. At each planning iteration, each agent plans $L = 10$ steps ahead. Single-agent budgets are 8 planning iterations for Gibson, HM3D-small, and space, and 20 for HM3D-large, chosen as the minimum needed for the best-performing methods to achieve reasonable reconstruction quality. Multi-agent experiments use three agents initialized at the same position to mimic practical deployment, each with half the single-agent budget.

b) Baselines: We compare our method against FisherRF and its batched variant, FisherRF-B [9], as well as VIMC [10], MAGICIAN [26], COVER [11], and GAVIS [6]. MAGICIAN serves as a coverage-based heuristic baseline, scoring trajectories by accumulated surface coverage gain. All methods use the same active mapping pipeline and sampler, without additional planning heuristics; each method selects the highest-scoring of 1,000 candidate trajectories. The sampler uses an RRT to generate collision-free trajectories with forward-facing poses indoors and smooth spline trajectories around spacecraft in the space scenes, with poses uniformly spaced in time. We extend FisherRF-B and MAGICIAN, which support batched-view evaluation, to multi-agent mapping, a setting not discussed in their original work.

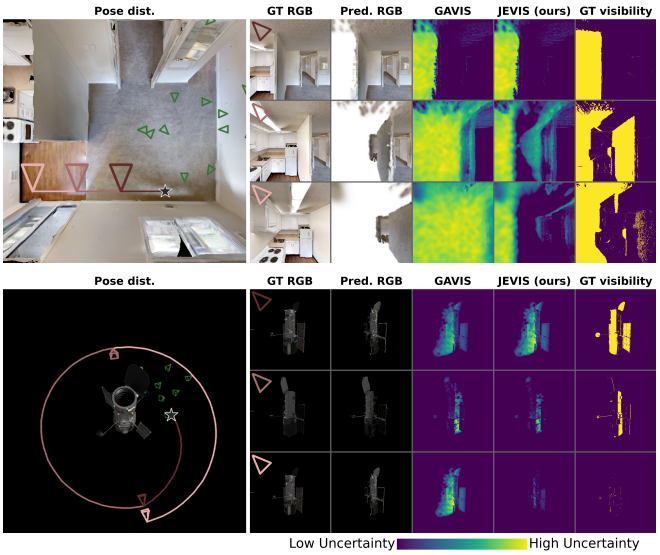


Fig. 3. **Qualitative uncertainty quantification.** JEVIS reduces estimated uncertainty in regions visible from preceding planned poses, whereas GAVIS scores each view independently and does not account for correlations between planned views. Top: indoor scene, where the candidate trajectory extends from observed into unseen regions with substantial view overlap. Bottom: space scene, where the candidate trajectory circles the spacecraft and revisits an earlier viewpoint. Within each scene, rows show successive candidate poses. GT Vis. denotes ground-truth mesh visibility, with bright regions indicating surfaces unseen by training views (green frustum) and preceding trajectory poses. In the shown examples, JEVIS's uncertainty maps more closely match GT Vis. than GAVIS's.

c) Metrics: We evaluate rendering quality with PSNR, SSIM, and LPIPS [35], and reconstruction quality with completion ratio (CR) [36] and visual coverage (VIS) [5]. Dataset averages include three runs for every scene.

B. Core Results

a) Joint Uncertainty Quantification (Q1): We evaluate joint uncertainty on 8 representative scenes using the area under the sparsification error curve (AUSE), extending single-view depth-based AUSE (AUSE-D) [37] and GT visibility-based AUSE (AUSE-V) [6] to complete trajectories (Tab. I). At each pose, the AUSE-D oracle is the absolute depth error after refitting the 3DGS model on the training views and preceding real observations; the AUSE-V oracle marks newly visible surfaces. Uncertainty maps are predicted from the initial reconstruction, while both oracle targets account for preceding observations. We pool pixels across all poses of each trajectory before computing AUSE, assessing uncertainty rankings both within and across views.

TABLE I. **Uncertainty quantification.** JEVIS achieves the best depth-based and visibility-based AUSE among the compared methods. Best results are **bolded** and second-best are underlined.

	FisherRF	FisherRF-B	VIMC	MAGICIAN	COVER	GAVIS	JEVIS (ours)
AUSE-D ↓	0.482	0.506	0.540	0.473	0.407	0.288	**0.144**
AUSE-V ↓	0.510	0.523	0.541	0.543	0.432	0.292	**0.188**

TABLE II. **Computational efficiency.** Mean per-view uncertainty quantification time on an NVIDIA A40 GPU.

	FisherRF	FisherRF-B	VIMC	MAGICIAN	COVER	GAVIS	JEVIS (ours)
Time/view (ms) ↓	15.82	16.30	13.80	38.72	10.69	**5.54**	7.12

TABLE III. **Long-horizon single-agent active mapping.** JEVIS achieves the best performance across all four datasets and all five metrics.

Method	Gibson					HM3D-small					HM3D-large					Space				
	PSNR ↑	SSIM ↑	LPIPS ↓	CR ↑	VIS ↑	PSNR ↑	SSIM ↑	LPIPS ↓	CR ↑	VIS ↑	PSNR ↑	SSIM ↑	LPIPS ↓	CR ↑	VIS ↑	PSNR ↑	SSIM ↑	LPIPS ↓	CR ↑	VIS ↑
FisherRF	18.94	0.740	0.412	0.347	0.402	16.19	0.674	0.484	0.221	0.306	15.45	0.641	0.499	0.128	0.255	16.71	0.608	0.375	0.402	0.382
FisherRF-B	19.15	0.739	0.407	0.345	0.397	16.54	0.684	0.474	0.230	0.309	16.19	0.655	0.483	0.151	0.306	17.02	0.612	0.372	0.431	0.401
VIMC	20.01	0.751	0.401	0.350	0.554	19.24	0.733	0.428	0.375	0.545	16.71	0.667	0.495	0.171	0.441	15.75	0.554	0.415	0.327	0.373
MAGICIAN	23.30	0.802	0.341	0.610	0.685	20.39	0.746	0.392	0.464	0.556	17.92	0.689	0.448	0.259	0.483	15.84	0.594	0.380	0.361	0.296
COVER	22.64	0.792	0.355	0.551	0.628	20.47	0.749	0.397	0.470	0.592	18.30	0.705	0.436	0.256	0.478	17.35	0.630	0.363	0.418	0.415
GAVIS	23.49	0.801	0.342	0.674	0.740	21.37	0.757	0.380	0.540	0.648	19.49	0.712	0.425	0.348	0.603	18.65	0.669	0.343	0.506	0.484
JEVIS (ours)	**25.25**	**0.832**	**0.312**	**0.766**	**0.835**	**22.56**	**0.777**	**0.358**	**0.599**	**0.735**	**20.12**	**0.721**	**0.418**	**0.369**	**0.689**	**19.39**	**0.694**	**0.332**	**0.523**	**0.545**

TABLE IV. **Long-horizon multi-agent active mapping.** JEVIS achieves the best performance across all four datasets and all five metrics with three agents. Each agent is allotted half the planning budget of the corresponding single-agent experiment.

Method	Gibson					HM3D-small					HM3D-large					Space				
	PSNR ↑	SSIM ↑	LPIPS ↓	CR ↑	VIS ↑	PSNR ↑	SSIM ↑	LPIPS ↓	CR ↑	VIS ↑	PSNR ↑	SSIM ↑	LPIPS ↓	CR ↑	VIS ↑	PSNR ↑	SSIM ↑	LPIPS ↓	CR ↑	VIS ↑
FisherRF	19.45	0.748	0.401	0.353	0.409	16.35	0.680	0.483	0.219	0.297	16.38	0.656	0.485	0.138	0.302	17.01	0.611	0.369	0.420	0.435
FisherRF-B	19.67	0.752	0.398	0.376	0.438	17.22	0.695	0.468	0.258	0.352	17.57	0.685	0.459	0.184	0.388	17.72	0.628	0.360	0.456	0.492
VIMC	19.57	0.748	0.406	0.333	0.524	19.09	0.729	0.433	0.343	0.516	17.47	0.673	0.488	0.191	0.488	16.44	0.569	0.405	0.367	0.436
MAGICIAN	24.56	0.821	0.316	0.658	0.737	21.68	0.767	0.378	0.507	0.600	19.19	0.714	0.426	0.282	0.553	16.71	0.615	0.368	0.387	0.344
COVER	22.03	0.785	0.360	0.491	0.573	20.76	0.757	0.396	0.463	0.570	19.40	0.717	0.424	0.265	0.559	17.85	0.635	0.355	0.456	0.498
GAVIS	24.01	0.809	0.330	0.653	0.737	21.48	0.763	0.382	0.521	0.625	20.01	0.724	0.413	0.326	0.618	19.27	0.694	0.331	0.491	0.523
JEVIS (ours)	**25.64**	**0.834**	**0.308**	**0.764**	**0.844**	**23.20**	**0.786**	**0.363**	**0.613**	**0.754**	**21.46**	**0.740**	**0.402**	**0.417**	**0.777**	**20.29**	**0.719**	**0.318**	**0.538**	**0.619**

JEVIS achieves the lowest AUSE-D and AUSE-V (Tab. I). We attribute this improvement to cross-view conditioning, which reduces the estimated uncertainty of regions unseen in the training views but covered by preceding planned views.

The qualitative results illustrate how JEVIS discounts shared uncertainty across planned views, producing conditional entropy maps that better match ground-truth visibility (Fig. 3). At the first candidate pose, both methods assign high uncertainty to regions unseen from the training views. When subsequent planned views reobserve these regions, JEVIS reduces their estimated uncertainty using visibility from preceding planned poses, whereas GAVIS continues to evaluate each view using only the training set.

Following GAVIS [6], we report the mean per-view uncertainty quantification time in Tab. II, measured as the time required to score a candidate trajectory divided by the number of views it contains. MAGICIAN has the highest runtime due to neural network inference. JEVIS is approximately 28% slower than GAVIS due to its additional visibility-field updates, but remains faster than all other baselines. Further optimizing these updates is left for future work.

b) Long-Horizon Single-Agent Active Mapping (Q2): JEVIS leads across four datasets and five metrics (Tab. III). GAVIS generally ranks second, substantially outperforming FisherRF and VIMC, which lack explicit visibility modeling, consistent with prior findings [6]. FisherRF-B still struggles with uncertainty estimation in unseen regions despite batching. JEVIS improves reconstruction quality, completeness, and coverage over GAVIS, as shown in Fig. 4 (top).

c) Long-Horizon Multi-Agent Active Mapping (Q3): JEVIS leads across all four datasets and five metrics, generally widening its margin over the strongest baseline relative to single-agent mapping (Tab. IV). Although MAGICIAN can be extended to multi-agent mapping via its coverage heuristic for batch view selection, its weaker performance in complex scenes may reflect limited generalization of its pretrained occupancy network. As illustrated in Fig. 4 (bottom), JEVIS agents explore complementary regions, whereas GAVIS agents follow overlapping trajectories. We additionally perform ablation studies (Tab. V) to assess within-trajectory and inter-agent conditioning for multi-agent scenarios: Ind. views (GAVIS) omits both, while Ind. agents retains only the former. With both forms of conditioning, JEVIS outperforms both variants on all five metrics.

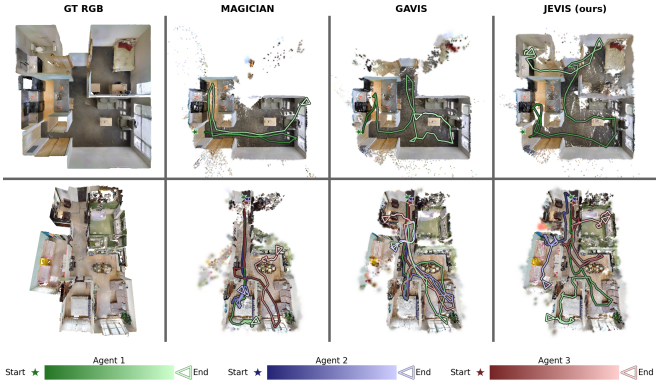


Fig. 4. **Qualitative active mapping.** JEVIS produces more complete reconstructions than GAVIS in the shown single-agent and multi-agent examples. Top: single-agent mapping on Gibson. Bottom: multi-agent mapping on HM3D-small, where JEVIS agents explore complementary regions while GAVIS agents follow largely overlapping trajectories. Colors distinguish agents; each trajectory runs from a star (start) to a frustum (end), with color progressing from dark to light over time.

TABLE V. **Ablation study in multi-agent active mapping.** Results are averaged over all scenes. *Ind. views* (GAVIS) estimates the entropy of each planned view independently; *Ind. agents* retains only within-trajectory conditioning. JEVIS uses both within-trajectory and inter-agent conditioning and achieves the best results on all five metrics.

	PSNR ↑	SSIM ↑	LPIPS ↓	CR ↑	VIS ↑
Ind. views (GAVIS)	21.32	0.751	0.366	0.498	0.633
Ind. agents	22.18	0.764	0.356	0.542	0.709
JEVIS	22.80	0.773	0.350	0.586	0.757

VI. Conclusion

We present Joint Entropy estimation in 3DGS via VISibility field (JEVIS), a principled framework for estimating the joint entropy of multiple observations of a 3DGS scene. By modeling uncertainty shared by planned views of common regions unseen in the training data via visibility fields, JEVIS extends uncertainty quantification from myopic next-best-view selection to long-horizon single- and multi-agent active mapping. JEVIS outperforms existing baselines across diverse indoor and space robotics environments. Building on

this formulation, future work could extend JEVIS to 3DGS-based active SLAM by jointly accounting for uncertainty in scene reconstruction and robot localization.

VII. Acknowledgments

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